

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 1 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 1

Subject Code & Name: 24AF1EMES104 - ENGINEERING MECHANICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 The primary objective of engineering analysis and design is to:

- a) Increase material usage
- b) Idealize real problems
- c) Achieve costlier solutions
- d) Avoid simplification

Composition of forces means:

- a) Replacing a force by two or more components
- b) Replacing multiple forces by their resultant
- c) Ignoring all forces
- d) Drawing the free body diagram

3 The algebraic sum of the moments of a force system about a point is equal to the moment of the resultant force about the same point. This statement is known as:

- a) Lami's theorem
- b) Varignon's theorem
- c) Pascal's theorem
- d) Newton's first law

4 For a body to be in static equilibrium, the following condition(s) must be satisfied:

- a) Sum of all forces must be zero
- b) Sum of all moments must be zero
- c) Both a and b
- d) Only sum of moments must be zero

5 According to Coulomb's laws of dry friction, the frictional force is:

- a) Proportional to the cube of normal reaction
- b) Always greater than normal reaction
- c) Independent of normal reaction
- d) Proportional to the normal reaction

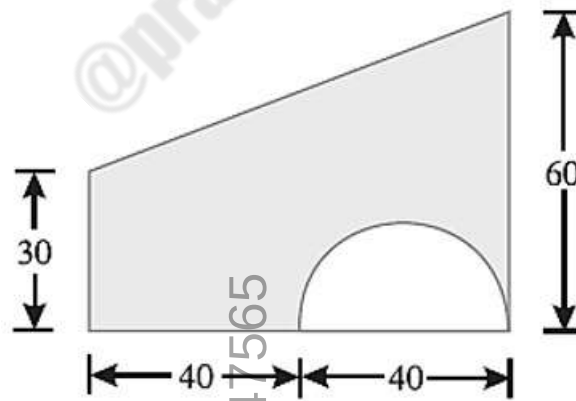
- 6 The condition of equilibrium for coplanar non-concurrent forces includes:
- Sum of horizontal components must be zero
 - Sum of vertical components must be zero
 - Sum of moments about any point must be zero
 - All of the above
- 7 Which of the following is not a type of load acting on beams?
- Point load
 - Uniformly distributed load (UDL)
 - Uniformly varying load (UVL)
 - Circular load
- 8 The method of sections is useful to:
- Find all member forces at once
 - Find forces in specific members quickly
 - Draw shear force diagram
 - Calculate moments of inertia
- 9 The first law of motion states that:
- $F = ma$
 - To every action there is an equal and opposite reaction
 - A body continues in its state of rest or uniform motion unless acted upon by an external force
 - Force is proportional to rate of change of momentum
- 10 Motion under gravity, neglecting air resistance, has:
- Constant acceleration upwards
 - Constant acceleration downwards equal to g
 - Variable acceleration
 - No acceleration
- In linear motion, the work-energy equation is expressed as:
- Work done = rate of change of acceleration
 - Work done = change in kinetic energy
 - Work done = change in momentum
 - Work done = constant
- 12 The product of force and velocity is called:
- Work
 - Power
 - Energy
 - Momentum

Q2. Solve the following.

- What are the types of loads and supports and explain them in details with neat sketches? 6
- The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force. 6

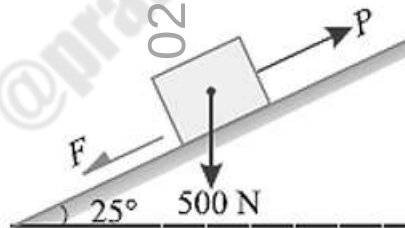
Q3. Solve the following.

- A) A semicircular area is removed from a trapezium as shown in figure (dimensions in mm) 6



Determine the centroid of the remaining area (shown hatched). Let left face and base of the trapezium be the axes of reference.

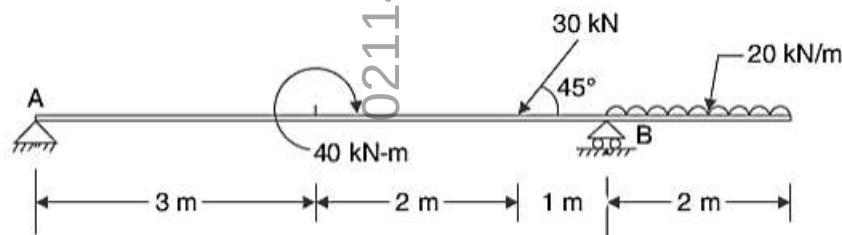
- B) A body of weight 500 N is lying on a rough plane inclined at an angle of 25° with the horizontal. It is supported by an effort (P) parallel to the plane as shown in figure. 6



Determine the minimum and maximum values of P, for which the equilibrium can exist, if the angle of friction is 20° .

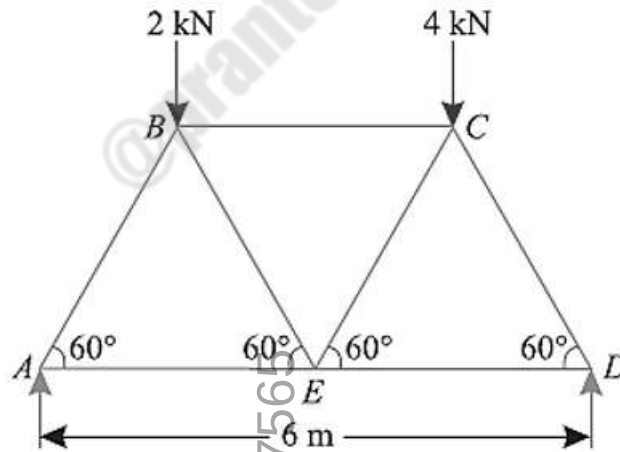
Q4. Solve Any Two of the following.

- A) Determine the reactions developed at supports A and B of overhanging beam shown in figure. 6



B)

6



Above figure shows a Warren girder consisting of seven members each of 3 m length freely supported at its end points. Find the forces in all the members of the girder, indicating whether the force is compressive or tensile.

What are the assumptions made, while finding out the forces in the various members of a framed structure?

6

Q5. Solve Any Two of the following.

A) A body moves along a straight line and its acceleration 'a' which varies with time is given by $a = 2 - 3t$. Five seconds after start of the observations, its velocity is found to be 20 m/sec. Ten seconds after start of the observation, the body is at 85 m from the origin. Determine its acceleration, velocity and distance from the origin.

B) Two ships leave a port at the same time. The first steams North-West at 32 kilometers per hour and the second 40° South of West at 24 kilometers per hour.
(a) What is the velocity of the second ship relative to the first in km per hour?
(b) After what time, they will be 160 km apart?

A body is projected at such an angle that the horizontal range is three times the greatest height. Find the angle of projection.

Q6. Solve Any Two of the following.

A) State and prove work energy principle.

B) Two weights 800 N and 200 N are connected by a thread and they move along a rough horizontal plane under the action of a force of 400 N applied to the 800 N weight as shown in figure. The coefficient of friction between the sliding surface of the weights and the plane is 0.3. Using D'Alembert's principle, determine the acceleration of the weight and tension in the thread.



Direct central impact occurs between a 300 N body moving to the right with a velocity of 6 m/s and 150 N body moving to the left with a velocity of 10 m/s. Find the velocity of each body after impact if the coefficient of restitution is 0.8.

*** End ***